

CS101 — Algorithms & Data Structures

Instructor: Dr. Maya Lindqvist

Term: Autumn 2026

Credits: 7.5 ECTS

This course is a first look at how to reason about running time, and how to pick a data structure without guessing. We start with asymptotic analysis, move through sorting and trees, and finish with graphs.

You will write small programs, but the point is not typing. The point is being able to say **why** an algorithm is fast enough, and what happens when the input doubles.

What you will be able to do

- Read a loop and give a tight big-O bound.
- Implement and compare insertion sort, mergesort, and quicksort.
- Use stacks, queues, hash tables, and binary heaps as tools rather than trivia.
- Run BFS, DFS, and Dijkstra on a weighted graph and explain the difference.

Bring a laptop to the lab sessions. Pair work is allowed on the assignment unless a brief says otherwise.

Weekly rhythm

Lectures on Tuesday. Labs on Thursday. Readings are short on purpose; do them before the lab or the exercises will feel like a riddle.

Office hours: Wednesday 13:00–14:30, room 4.12.

Assessment overview

The course is assessed with one programming assignment, a graph project checkpoint, and a written exam. Weights and dates are in the table below. Late work loses 10% per day unless you have agreed an extension in writing.

Keep the table. Almost every “when is it due?” question in this course is answered here.

Item	Weight	Due
Assignment 1: Sorting visualiser	20%	14 October 2026, 23:59
Graph project checkpoint	20%	28 November 2026, 23:59
Final exam	60%	12 December 2026, 09:00

The exam is closed book. You may bring one double-sided A4 sheet of handwritten notes. Calculators are not useful and are not allowed.

If a date in a later lecture note disagrees with this table, **this table wins**. Email the teaching team if you spot a conflict.